v4 Cheatsheet Evaluation

All Models x Hard Datasets

SAIR Mathematics Distillation Challenge

Equational Theories Stage 1

Prompt: v4_10KB_cheatsheet.md

2026-04-06

4,683 API calls | 669 problems/model | 7 models | ~$2.50 total cost

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Model	hard1 (69)	hard2 (200)	hard3 (400)	Parse Err	Bias Pattern
grok-4.1-fast	**53/68 (77.9%)**	**153/200 (76.5%)**	**242/400 (60.5%)**	1	Balanced
gemma-4-31b	**43/69 (62.3%)**	**145/197 (73.6%)**	**202/385 (52.5%)**	18	TRUE bias (confab)
gemini-flash-lite	**51/69 (73.9%)**	**104/200 (52.0%)**	**225/400 (56.2%)**	0	FALSE bias
deepseek-v3.2	**45/69 (65.2%)**	**101/200 (50.5%)**	**206/400 (51.5%)**	0	Total FALSE (0% TRUE recall)
llama-3.3-70b	**45/69 (65.2%)**	**100/200 (50.0%)**	**206/400 (51.5%)**	0	Total FALSE (0% TRUE recall)
gpt-oss-20b	**38/65 (58.5%)**	**76/174 (43.7%)**	**179/358 (50.0%)**	72	TRUE bias + parse err
gpt-oss-120b	**29/66 (43.9%)**	**102/194 (52.6%)**	**170/369 (46.1%)**	40	Extreme TRUE bias

4,683 API calls | 669 problems/model | Total cost: ~$2.50 | Datasets: hard1 (69), hard2 (200), hard3 (400)

Color key: **≥ 70%** (green) | **55–69%** (gold) | **<55%** (red) Bias: Balanced (green) | FALSE bias (grey) | TRUE bias (red)

Insights & Recommendations

KEY INSIGHTS

(from JSON analysis + community intel)

CRITICAL **1024-token cap is the #1 GPT-OSS failure driver**

GPT-OSS-120b: ~90% of failures hit exactly 1024 tokens mid-reasoning. No VERDICT emitted; harness defaults unpredictably. Community best practice: emit VERDICT as first output line, reason after.

CRITICAL **v4 creates extreme polarization across model families**

DeepSeek/Llama/Gemini: 0–9% TRUE recall (always FALSE). GPT-OSS: 85–91% TRUE recall (always TRUE). Only Grok stays balanced. The step-by-step decision procedure is too rigid for weaker models and too loose for stronger ones.

HIGH **Betka's 98% sheet uses contradiction motifs C1–C14**

Our v4 motifs over-trigger on FALSE problems (59 confabs in Grok alone). Betka's sheet forces feature computation (rhsVars, topShape, Lx/Rx, xTop) before any rule fires, with hard-stop on first match.

HIGH **Llama/DeepSeek treat cheatsheet as pure lookup table**

317/318 Llama failures are identical 'RULE: default false' boilerplate (median 88 tokens). No reasoning attempted. DeepSeek fabricates counterexamples (38%) when no rule matches.

Gemini hallucinates structural rules not in the cheatsheet

Invents 'rightmost exclusion' and 'count exclusion' heuristics (53% of failures). Pattern-matches the cheatsheet's naming convention to fabricate plausible but invalid rules.

NEXT STEPS

(prioritized)

1. **CRITICAL** **VERDICT-first output format**

   Emit 'VERDICT: TRUE' or 'VERDICT: FALSE' as the first line, then reasoning. Eliminates all token-truncation parse errors (72 on GPT-20b, 40 on GPT-120b). Community consensus (Betka, Heath, stokarz).

2. **CRITICAL** **Counterexample hard-stop with named rule lock**

   Once a counterexample is found: cite rule, output FALSE immediately, stop reasoning. GPT-120b finds correct counterexamples then 'talks itself out of it' with fluent wrong proofs. Prevents 288 GPT-120b confabulations.

3. **HIGH** **Add Spine Isolation Theorem (McKenna, EQT01-000011)**

   One sentence: 'A pure left-spine equation can only imply equations that are themselves pure left-spine of equal or greater depth.' Validated on 1.54M pairs with zero exceptions. Fixes +32 hard3 problems.

4. **HIGH** **Adopt Betka's feature-first protocol for TRUE routes**

   Compute source features (rhsVars, topShape, Lx, Rx, xTop, square) BEFORE checking motifs. Our v4 lets models skip feature computation. Betka achieves 98% on hard200 with GPT-120b.

5. **MEDIUM** **Force minimal reasoning before default-FALSE**

   Add: 'Before defaulting to FALSE, attempt one substitution x=y=z and check if Eq2 follows from Eq1.' Llama outputs 'default false' in 88 tokens with zero analysis.

6. **MEDIUM** **Increase max_tokens to 2048 for GPT-OSS models**

   GPT-120b correct answers average 1063–1188 tokens. Current 1024 cap truncates ~90% of failures. Doubling to 2048 costs $0.10 more per run.

Sources: JSON reasoning analysis | Betka (EQT01-000014, 98% hard200) | McKenna (Spine Isolation, +32 hard3) | Tao (50/50 equivalence/non-equivalence, 109/200 hard200)

Appendix: Grok 4.1 Fast

v4_10KB_cheatsheet.md | Backend: OpenRouter | Model: grok-4.1-fast

Dataset	Accuracy	TRUE→TRUE	TRUE→FALSE	FALSE→TRUE	FALSE→FALSE	Parse Err
hard1	77.9%	16/24	8	7	37	1
hard2	76.5%	75/100	25	22	78	0
hard3	60.5%	102/195	93	65	140	0

Most balanced model — 67% TRUE recall on hard1, 75% on hard2. Only model above 60% on hard3.

FAILURE ANALYSIS (from JSON reasoning)

Failure Pattern	Count	Detail
Default FALSE fallback (no TRUE route fired)	**126 (57%)**	Cheatsheet TRUE-detection rules too narrow; model correctly follows protocol but defaults to FALSE on unrecognized TRUE patterns.
Contradiction motif over-trigger	**59 (27%)**	C8/C9/C13 motifs (rhsVars, Lx/Rx flags, topShape) fire on FALSE problems. Motifs derived from biased sample; structural match doesn't guarantee implication.
Constant-operation rule misfire	**19 (9%)**	Disjoint variable sets on Eq1 sides wrongly triggers 'constant magma' conclusion. Rule is mathematically overbroad.
Projection rule misidentification	**10 (5%)**	Left/right projection rule fires but T actually contains x, or derivation from projection to Eq2 is incorrect.

Appendix: Gemma 4 31B

v4_10KB_cheatsheet.md | Backend: OpenRouter | Model: gemma-4-31b

Dataset	Accuracy	TRUE→TRUE	TRUE→FALSE	FALSE→TRUE	FALSE→FALSE	Parse Err
hard1	62.3%	21/24	3	23	22	0
hard2	73.6%	85/97	12	40	60	3
hard3	52.5%	149/189	40	143	53	15

Strong TRUE recall (88% on hard1/hard2) but heavy confabulation on FALSE problems. 18 parse errors (empty API responses).

FAILURE ANALYSIS (from JSON reasoning)

Failure Pattern	Count	Detail
Contradiction motif misfire (FALSE→TRUE)	**~153 (55%)**	Model walks through motif checklist, fails to match, then reverses mid-reasoning to output TRUE. Confabulation at the rule-application layer.
Left projection misfire (FALSE→TRUE)	**~43 (15%)**	Claims 'left projection' rule applies but structural conditions don't hold. Pattern-matches rule name without verifying preconditions.
Right-zero witness misuse (TRUE→FALSE)	**35 (13%)**	When Eq1 fails in right-zero magma, concludes FALSE. Logic error: a magma where Eq1 fails is NOT a valid counterexample.
Syntactic exclusion misfires (TRUE→FALSE)	**15 (5%)**	Leftmost/count exclusion rules applied without verifying preconditions. Pattern-matching syntax without understanding semantics.
Parse errors (empty API response)	**18 (6%)**	output_tokens=0, empty content. API-level failures/timeouts, not model reasoning issues.

Appendix: Gemini 3.1 Flash Lite

v4_10KB_cheatsheet.md | Backend: OpenRouter | Model: gemini-flash-lite

Dataset	Accuracy	TRUE→TRUE	TRUE→FALSE	FALSE→TRUE	FALSE→FALSE	Parse Err
hard1	73.9%	9/24	15	3	42	0
hard2	52.0%	9/100	91	5	95	0
hard3	56.2%	36/195	159	16	189	0

Massive FALSE bias — 9% TRUE recall on hard2. Near-perfect FALSE accuracy but misses almost all TRUE implications. Zero parse errors.

FAILURE ANALYSIS (from JSON reasoning)

Failure Pattern	Count	Detail
Fabricated rightmost exclusion heuristic	**154 (53%)**	Model invents a 'rightmost variable must match' rule not in the cheatsheet. Fires on ~58% of TRUE problems. Hallucinated structural heuristic.
Default FALSE on non-bare equations	**64 (22%)**	When Eq1 is not 'x = T' form, immediately concludes no TRUE routes apply. Skips all non-trivial reasoning.
Fabricated leftmost exclusion heuristic	**35 (12%)**	Symmetric to rightmost: checks leftmost variable preservation and rejects TRUE implications. Another hallucinated rule.
Contradiction motif over-firing (FALSE→TRUE)	**15 (5%)**	Incorrectly claims equation 'collapses' magma. Accounts for 63% of the 24 false positives.
Count exclusion misapplication	**12 (4%)**	Claims variable occurrence counts must be preserved across implication. Not a valid algebraic invariant.

Appendix: DeepSeek V3.2

v4_10KB_cheatsheet.md | Backend: OpenRouter | Model: deepseek-v3.2

Dataset	Accuracy	TRUE→TRUE	TRUE→FALSE	FALSE→TRUE	FALSE→FALSE	Parse Err
hard1	65.2%	0/24	24	0	45	0
hard2	50.5%	1/100	99	0	100	0
hard3	51.5%	2/195	193	1	204	0

Total FALSE bias — 0–1% TRUE recall. Perfect FALSE detection but no signal on TRUE. Zero parse errors.

FAILURE ANALYSIS (from JSON reasoning)

Failure Pattern	Count	Detail
Step 6 'default FALSE' fallback	**165 (52%)**	No TRUE route fires; model follows cheatsheet correctly and defaults to FALSE. TRUE-detection rules have zero coverage on hard problems.
Fabricated / incorrect counterexamples	**121 (38%)**	Constructs small magma, claims Eq1 holds and Eq2 fails, but math is wrong — Eq1 often fails in that magma too. Not a valid counterexample.
Wrong projection rule application	**~20 (6%)**	Correctly identifies left-zero/right-zero behavior for Eq1, then mis-reduces Eq2 with flawed variable trace algebra.
Wrong parity/syntactic exclusion	**~10 (3%)**	Applies parity exclusion rule with incorrect parity computation. 'Left side has one x, right has two' — miscounted.

Appendix: Llama 3.3 70B

v4_10KB_cheatsheet.md | Backend: OpenRouter | Model: llama-3.3-70b

Dataset	Accuracy	TRUE→TRUE	TRUE→FALSE	FALSE→TRUE	FALSE→FALSE	Parse Err
hard1	65.2%	0/24	24	0	45	0
hard2	50.0%	0/100	100	0	100	0
hard3	51.5%	2/195	193	1	204	0

Total FALSE bias — 0% TRUE recall on hard1/hard2. Treats cheatsheet as pure lookup table with no generalization.

FAILURE ANALYSIS (from JSON reasoning)

Failure Pattern	Count	Detail
'RULE: default false' on every unmatched problem	**317 (99.7%)**	Identical boilerplate on all failures: 'The given equations do not match any exact TRUE routes.' Median 88 tokens output. Zero algebraic reasoning attempted.
No chain-of-thought reasoning	**317 (100%)**	Every response is a direct verdict + one-sentence justification. Never attempts substitution, variable analysis, or counterexample construction.
Only 2 TRUE problems matched (left projection)	**2 correct**	hard3_0002 and hard3_0005 matched the 'left projection' named rule exactly. Everything else falls through to default FALSE.

Appendix: GPT-OSS 20B

v4_10KB_cheatsheet.md | Backend: OpenRouter | Model: gpt-oss-20b

Dataset	Accuracy	TRUE→TRUE	TRUE→FALSE	FALSE→TRUE	FALSE→FALSE	Parse Err
hard1	58.5%	16/23	7	20	22	4
hard2	43.7%	53/89	36	62	23	26
hard3	50.0%	130/174	44	135	49	42

TRUE bias + 72 parse errors. Singleton collapse misfire is the dominant confabulation source.

FAILURE ANALYSIS (from JSON reasoning)

Failure Pattern	Count	Detail
Singleton/collapse rule misfire (FALSE→TRUE)	**217 (58%)**	Incorrectly triggers 'singleton collapse' heuristic. 183/217 contain the word 'singleton'. Fires on bare-variable left sides that superficially resemble TRUE patterns.
Token limit truncation (parse errors)	**72 (19%)**	All parse errors hit exactly 1024 output tokens. Model cut off mid-reasoning before emitting VERDICT line. Hard limit issue.
Parity exclusion misfire (TRUE→FALSE)	**74 (20%)**	Named 'parity exclusion' fires incorrectly. Model computes parity mismatch and concludes FALSE on genuinely TRUE implications.
Fabricated counterexample (TRUE→FALSE)	**31 (8%)**	Constructs left-zero or 2-element magma, checks Eq2 fails, declares FALSE — but either the check is wrong or Eq1 doesn't hold in that magma.

Appendix: GPT-OSS 120B

v4_10KB_cheatsheet.md | Backend: OpenRouter | Model: gpt-oss-120b

Dataset	Accuracy	TRUE→TRUE	TRUE→FALSE	FALSE→TRUE	FALSE→FALSE	Parse Err
hard1	43.9%	21/23	2	35	8	3
hard2	52.6%	87/99	12	80	15	6
hard3	46.1%	150/176	26	173	20	31

1024-token output cap is the root cause of ~90% of failures. Model needs more tokens to complete the decision procedure.

FAILURE ANALYSIS (from JSON reasoning)

Failure Pattern	Count	Detail
Token-limit truncation → silent TRUE default	**~288 (78%)**	Reasoning chain exhausts 1024-token budget mid-step. No VERDICT emitted; harness defaults to TRUE. hard1: 32/35 confab at limit, hard3: 165/173 at limit.
'Constant operation' rule misfire	**~13 (4%)**	Non-truncated false positive. Checks 'disjoint variable sets' but gets disjointness wrong, or Eq1 doesn't actually force a constant operation.
'Projection' rule misfire	**~6 (2%)**	Verifies only variable-non-overlap condition, ignoring structural requirements. Declares TRUE when preconditions aren't met.
Token-limit truncation → silent FALSE default	**~40 (11%)**	Mirror of #1: all TRUE→FALSE errors also hit 1024 tokens. Model never reaches TRUE conclusion before cutoff.
Parse errors (empty response)	**40 (11%)**	Empty reasoning + empty content, exactly 1024 tokens. Some are API timeouts (latency >1500s). Not malformed output — just never produced VERDICT.